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ORIGINAL ARTICLE

Effect of micro-oxygen pretreatment on gas production characteristics of anaerobic digestion of kitchen waste

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Abstract

Micro-oxygen pretreatment of kitchen waste was done to improve the gas production efficiency of anaerobic fermentation and shorten the fermentation cycle of kitchen waste. The results demonstrated that with an oxygen supply of 20 mL, the peak gas production increased from 498 to 601 mL, and the theoretical gas production increased by 4.5 mL/g VS (volatiles solid) compared with those of the control group R1–R4. The anaerobic fermentation entered the late stage on day 25, which was 13 days earlier than that of the control group. The chemical oxygen demand and ammonia nitrogen contents in the treatment groups of R1–R4 were relatively stable. An oxygen supply of 20 mL was conducive to stimulating the facultative anaerobes, promoting the decomposition of nitrogenous compounds and organic matter, and improving the anaerobic fermentation efficiency. A lesser amount of oxygen ($\leq$ 15 mL) was unable to effectively improve the gas production efficiency of anaerobic fermentation, although it exerted a certain inhibitory effect on anaerobic fermentation.

Keywords Micro-oxygen pretreatment · Kitchen waste · Gas production characteristics · Anaerobic digestion

Introduction

With the continuous improvement in people's living standards in China, the output of kitchen waste has also gradually increased. Kitchen waste has brought serious environmental pollution issues and a huge waste of resources. As a utilizable resource with rich organic matter (OM), kitchen waste is gaining more and more attention [1, 2]. Anaerobic fermentation is an approach to achieve reuse, reduce and recycle. The reaction is mainly divided into three stages, namely the hydrolytic acidification stage, the hydrogen production acetic acid stage, and the methanogenesis stage. In the hydrolytic acidification stage OM is converted into small molecule organic matter by microorganisms. The small molecule organic matter is then converted into volatile fatty acids by the hydrogen-producing and acetic acid-producing groups of bacteria, and finally CH_4 is produced by the methanogens using the volatile fatty acids. Anaerobic fermentation of kitchen waste, can not only reduce the OM content in kitchen waste, but also produce methane as a clean energy source. In addition, the produced biogas residue can also be used as a fertilizer, truly turning kitchen waste into a biomass resource [3, 4]. However, a major shortcoming is a long initial stage of the kitchen waste anaerobic fermentation. Thus, carrying out efficient anaerobic digestion for kitchen waste has become a major concern of researchers.

In the progress in research on organic matter anaerobic digestion pretreatment has attracted increasing attention in the last decade. Studies have shown that the commonly used pretreatment methods for anaerobic digestion of organic matters mainly include mechanical (ultrasonic [5, 6], microwave [7], electromagnetic high pressure homogenization

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[8], etc.) pretreatment, thermal pretreatment [9], chemical (acid [10], alkali [11], ozone [12], Fenton [13], Fe^{2+} ion-activated persulfate oxidation [14, 15]) pretreatment, and biological (fungi [16], enzymes [17], etc.) pretreatment.The pretreatment for anaerobic digestion of kitchen waste mainly aims to improve its rate of hydrolysis [18]. Araz Sedqi, et al. conducted aerobic pretreatment of the kitchen waste with four minerals (bentonite, fly ash, lightly burned MgO, and dolomite powder) for 4 h, and discovered that the alkaline minerals could adjust pH and enhance the capacity of ion exchange adsorption, thereby promoting the anaerobic fermentation of kitchen waste [19]. Zhang et al. pretreated the kitchen waste with ultrasound to boost the yield of volatile fatty acids (VFAs), and found a 27.2% increase in their yield from kitchen waste after sonication [20]. Further, the degradation efficiency could be enhanced by homogenizing the solid waste or increasing the specific surface area of solid particles to make it conducive to improving heat and mass transfer of the matrix [21, 22]. Li et al. found that thermal pretreatment reduced the oil potential, increased the stagnation period of proteins and oils, but increased the cumulative methane production. However, these pretreatments still have the problem of insufficient utilization of matrix [9]. The combined fermentation process of producing ethanol and methane from kitchen waste can overcome the problem of insufficient utilization of matrix and improve the energy recovery of the subsequent methane fermentation system. Li et al. compared the performance of micro-aerobic and anaerobic fermentation with corn stalk as substrate, and found that micro-aerobic conditions have strong buffer capacity and can tolerate the accumulation of large volatile fatty acids [23]. Martı´nez-Valdez et al. studied the mixing effect of aerobic pretreated organic fraction of municipal solid waste (OFMSW) and untreated OFMSW in the process of anaerobic digestion, and found that adding OFMSW after 5 days of pretreatment to untreated OFMSW could increase methane production by 18%, and that there was no significant difference in the lag time of methanogenic phase [24]. The effects of aerobic pretreatment and adding fly ash on methane production of mixed sludge were studied by Huilinir and Montalvo [25, 26], who found that the combined use of aerobic pretreatment and fly ash could increase methane production by more than 70%. Subsequently, Rafieenia et al. carried out anaerobic digestion of three substrates rich in carbohydrates, proteins, and lipids after aerobic pretreatment, respectively, and observed that aerobic pretreatment could be used to improve substrate conversion efficiency and CH_4 production in two-stage anaerobic digestion process [27]. In this study, we explored the effect of the micro-oxygen environment on the gas production characteristics of kitchen waste to improve the gas production efficiency during anaerobic digestion. By adding different amounts of oxygen to different experimental groups, we analyzed the changes in various parameters during the fermentation process, studied the influence mechanism on each parameter, and tried to improve the fermentation efficiency. This study provides theoretical guidance for a practical set up of anaerobic fermentation of kitchen waste.

Materials and methods

Experimental materials

The kitchen waste used in this study was obtained from the Southern District Food Court of Shenyang Aerospace University. The hard-to-decompose materials such as eggshells, bones, and bags, etc. were removed because they cannot be anaerobically fermented. The inoculated sludge was obtained from the concentrated pool of the sewage treatment plant in northern Shenyang and was domesticated for 48 h after retrieving. The properties of the kitchen waste and the inoculated sludge are mentioned in Table 1.

Experimental methods

The sequencing batch reactor (SBR) for anaerobic fermentation comprised five units: a temperature controller, a reaction tank, a gas collecting unit, a drainage collecting unit and a connecting unit (Fig. 1). All units were connected to form a through-type closed equipment, where the reaction tank, the gas collecting unit, and the drainage collecting unit, 20 cm in height and all had an effective volume of 1 L. The entire small anaerobic fermentation bioreactor was strictly controlled and airtight to ensure an anaerobic environment. At the start of the experiment, 100 g kitchen waste and 300 g inoculated sludge was added, and the volume was set to 1 L with deionized water. A total of five experimental groups was set up. In the treatment groups R1–R4, oxygen at 5 mL, 10 mL, 15 mL, and 20 mL was supplied respectively, and the group R5 was set as the blank control with no external oxygen supply. The experimental temperature was set at 37 °C. The variation patterns of primary parameters such as daily gas production, cumulative gas production, pH, COD, and ammonia nitrogen were evaluated. The detection methods, instrument models, and measurement accuracy of these parameters are shown in Table 2.

Table 1 Properties of kitchen waste and inoculated sludge

Item	Total solid content TS (%)	Volatile solid content VS (%)
Kitchen waste	21.78	19.53
Inoculated sludge	30.63	18.06

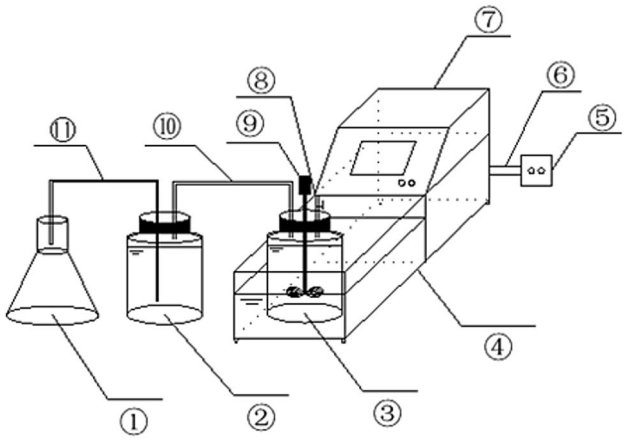


Fig. 1 An anaerobic fermentation reactor with micro-oxygen pretreatment. (1) Drainage collecting unit; (2) Gas collecting unit; (3) Anaerobic digestion reaction unit; (4, 5, 6, 7) Thermostatic water bath controller; (8) Sampling port and oxygen supply port; (9) Timing automatic mixer; (10, 11) Connecting units

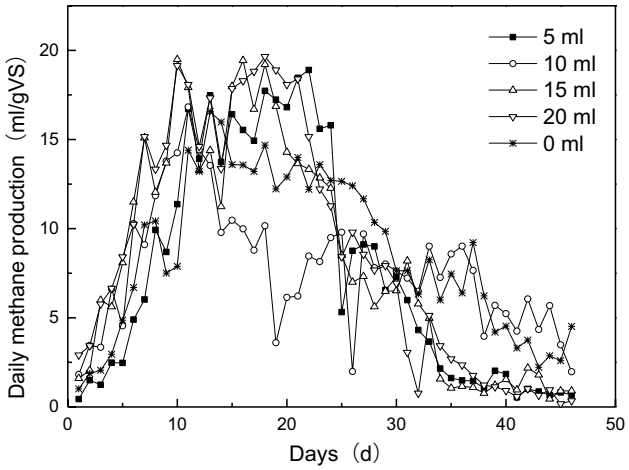


Fig. 2 Variations in daily methane yield in the micro-oxygen pretreatment experiment. [Time(d), Daily methane yield]

Results and discussion

Effect of oxygen concentration on the characteristics of gas production

The variations in daily and cumulative methane yields of the anaerobic digestion of kitchen waste at varying concentrations of micro-oxygen a period of for 50 days are shown in Figs. 2, 3. All treated and control groups exhibited low methane yields initially. This was because, at the early stage of fermentation, the microbes in the inoculated sludge were still in the adaptation phase, and the hydrolytic fermenting microbes were the ones working primarily [28]. The theoretical gas production can be obtained as the ratio of daily gas production to the VS content of the raw material. With the decomposition of organic macromolecules and the improved activity of methanogens, the groups R1–R4 reached the peak of first methane production at around day 10, Methane concentrations were 59.00%, 62.00%, 58.00%, 59.00%, and 60.00%, and the calculated theoretical methane yields were 15.43 mL/gVS(5 mL),15.56 mL/gVS(10 mL),18.03 mL/gVS(15 mL) and 18.00 mL/gVS(20 mL). Whereas the control group reached the first gas peak yield on day 14, with a theoretical methane yield of 25.50 mL/gVS. As time progressed, the methane yield exhibited varying patterns with respect to various micro-oxygen concentrations. As shown in Fig. 2, the rise in methane yields of R1–R4 was faster with higher peak values than that in the control group R5, indicating improved gas production efficiency of the entire fermentation cycle. The treatment groups R1–R4 entered the late stage of the experiment by the 25th day. Around this time, the daily gas production was in a declining stage and the hydraulic retention time reached 25 days. This indicates the decomposition of most of the organic matter in the reaction process is decomposed and the gas production starts to decline, while the control group R5 had a longer fermentation cycle and entered the late fermentation stage only on

Table 2 Test parameter detection method

Parameter	Detection method	Instrument and model	Instrument manufacturers	Measurement accuracy
Methane production	Drainage method, methane concentration analyzer	Methane concentration online analyzer (GT901)	Shenzhen Korno Electronic Technology Co., Ltd	0.1%
pH	Portable pH meter	Lei-ci pH meter (PHB-4)	Shanghai Yidian Scientific Instrument Co., Ltd	0.01
COD	Potassium dichromate method	COD Analyzer (CTL-12)	Chengde Huatong Environmental Protection Instrument Factory	0.1 mg/L
NH_4^+-N	Nessler's reagent spectrophotometry	Lei-ci Multi-parameter analyzer (DGB-401)	Shanghai Yidian Scientific Instrument Co., Ltd	0.01 mg/L

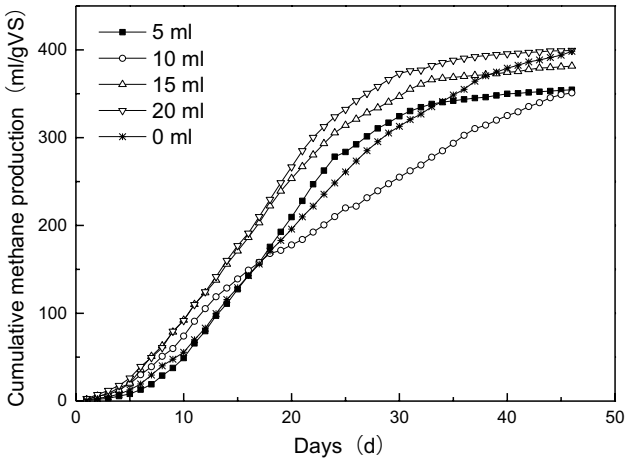


Fig. 3 Variations in cumulative methane yield in the micro-oxygen pretreatment experiment. [Time(d), Cumulative methane yield]

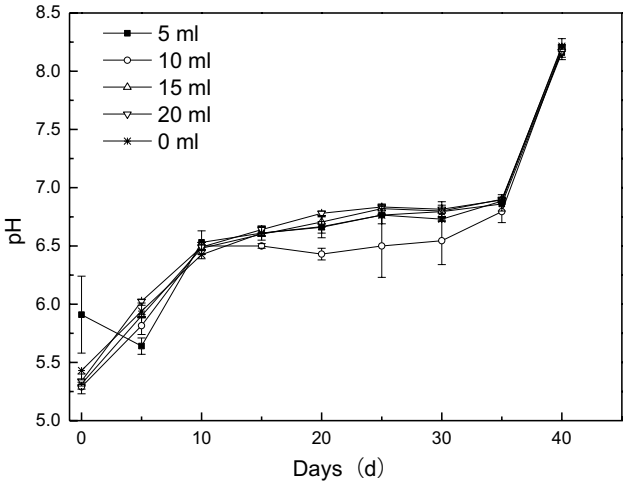


Fig. 4 pH variations regarding different oxygen concentrations. (R1—Pass 5 ml of oxygen; R2—Pass 10 ml of oxygen; R3—Pass 15 ml of oxygen; R4—Pass 20 ml of oxygen; R5—Pass 0 ml of oxygen;)

day 38. In inter-group comparisons of R1–R4, it was found that with the increase in oxygen supply, and the peak value of methane yield gradually increased, along with a gradual improvement in the efficiency of anaerobic fermentation. Nevertheless, the treatment groups were unstable during the peak gas production stage, probably due to an unstable oxygen supply. Feng et al. [29] demonstrated that the micro-oxygen amount could optimize the anaerobic fermentation system and increase the rate of fermentation by controlling the oxygen content under the initial conditions of anaerobic digestion of kitchen waste. Cumulative methanogenesis is consistent with the modified first-order hydrolysis model and the modified Gompertz model. This study uses a modified Gompertz model to fit the cumulative gas production. The fitting curve is shown in Fig. 3. The R1–R5 fitting curves R^2 are all above 0.99. The fitting effect is good. It can be seen that the modified Gompertz model can reflect the change of the cumulative gas production of anaerobic fermentation. The trends of cumulative methane yield for all groups was similar (Fig. 3), among which the curves of R3 (15 mL oxygen) and R4 (20 mL oxygen) exhibited with the highest slopes and superior gas production efficiencies. As the hydrolysis and acidification stage is the rate-limiting stage of anaerobic digestion, in the early hydrolysis stage of the anaerobic digestion, the addition of trace oxygen promotes the increase in numbers of facultative anaerobic bacteria such as yeast and Escherichia coli and is beneficial to the hydrolysis of macromolecular organic matter, Among the five groups, the highest cumulative methane yield was observed in R5. The final cumulative methane yields of R1–R5 were respectively 354.51 mL/gVS, 350.96 mL/gVS, 381.56 mL/gVS, 399.25 mL/gVS, 398.44 mL/gVS. The above results indicate that the introduction of trace oxygen can shorten the reaction cycle of the hydrolysis stage of the anaerobic fermentation, improve the high anaerobic fermentation treatment efficiency of the hydrolysis and acidification stage, and shorten the reaction cycle of anaerobic fermentation. Further, with the increase in the amount of oxygen, there is a gradual increase in efficiency.

Effect of oxygen concentration on pH of the reaction system

The variation in the pH of each group at different oxygen concentrations is shown in Fig. 4. In the initial stage, the anaerobic fermentation underwent hydrolysis/acidification. During this stage, the decomposition of organic macromolecules like proteins and fats occurred in the reaction tank to form small organic molecules like fatty acids, producing a large amount of free H^+, which led to drop in pH in the reaction tank [30–32]. As the fermentation progressed, the methanogens gradually adapted to the environment in the tank and began to use the free H^+ in the system to synthesize methane and CO_2, resulting in an increase in pH of the system. On comparing the groups, R2 presented a pH significantly lower than that of the other groups during the enrire process, indicating a low efficiency of H^+ utilization by methanogens under such an oxygen concentration (10 mL). In the next stage, the pH of all groups increased sharply, indicating the depletion of free H^+ by methanogens in the reaction system. However, at this stage the rate of increase in pH was not much different across the groups. The above results demonstrate that the micro-oxygen environment may improve the efficiency of H^+ utilization by methanogens, promote the activity of facultative anaerobes,

and enhance the gas production efficiency of anaerobic fermentation without sabotaging the anaerobic environment.

Effect of oxygen concentration on ammonia nitrogen content in the reaction system

In anaerobic fermentation, ammonia nitrogen exists primarily in the forms of free ammonia (NH_3) and ammonium cation (NH_4^+). During the anaerobic fermentation, favorable ammonia nitrogen could be used as a nutrient to provide energy for the growth of microbes. However, the physiological activity of methanogens is inhibited by excessively high ammonia nitrogen, resulting in ammonia inhibition [33, 34]. The variation of ammonia nitrogen during the fermentation at different oxygen concentrations is shown in Fig. 5. In the initial stages of fermentation, the ammonia nitrogen content was low in all the groups. As the fermentation proceeded, the rate of ammonia nitrogen produced by hydrolysis was much higher than its consumption in the reaction system, causing significantly elevated ammonia nitrogen content in all the groups (R1–R5). The ammonia nitrogen contents of R1–R5 increased to 464 ± 26 mg/L, 462 ± 17 mg/L, 471 ± 32 mg/L, 483 ± 23 mg/L, and 465 ± 14 mg/L, respectively, all of which stabilized around day 5. During the stable stage from day 5 to day 30, the content of ammonia nitrogen was only slightly higher in the R4 group than in R1–R3 groups and was not significantly different among R1–R4. In uninduced ammonia nitrogen inhibition, higher ammonia nitrogen could provide abundant raw materials for microbial growth and metabolism, allowing methanogens to utilize more small organic molecules, thereby enhancing the methane yield [35, 36]. In the later stage of fermentation, most of OM was exhausted, resulting in a higher rate of ammonia nitrogen consumption than production; hence, the ammonia nitrogen concentration decreased. Studies have shown that the addition of different amounts of oxygen does not affect the performance of anaerobic digestion, but can improve the efficiency of desulfurization in anaerobic fermentation and increase the redox potential of the system [37, 38]. Therefore, it can be inferred that during the anaerobic fermentation of kitchen waste, the ammonia nitrogen content of the micro-oxygen pretreatment groups was higher than that of the untreated control group, and that it increased with the increase in oxygen supply. The above results showed that the facultative anaerobes improved their capacity to produce and utilize nitrogenous compounds. Furthermore, increasing oxygen content promoted the activities of facultative anaerobes in the system and enhanced the decomposition of organic macromolecules during fermentation.

Effect of oxygen concentration on COD content in the reaction system

The changes in COD directly represented the OM changes in the anaerobic fermentation system [39] (Fig. 6). In the initial stage of microbial adaptation, the degradation rate of small organic molecules was lower than that of larger ones, leading to a rise in cumulative COD in R1–R5. As the fermentation proceeded, the COD values of R1–R5 peaked on days 6–10, at 20,720 ± 220 mg/L, 31,110 ± 670 mg/L, 24,167 ± 390 mg/L, 34,410 ± 510 mg/L, and 26,820 ± 530 mg/L, respectively. The peak COD indicates the time when the microbes were the most active. The peak value and the time required to reach it may be used as a reference to determine the fermentation cycle. When the rate of microbial degradation of small organic molecules is faster than that of larger organic macromolecules, methanogens initiate the decomposition of substantially small organic

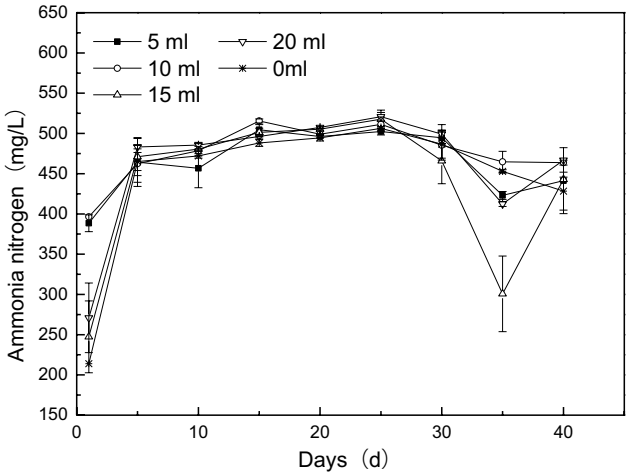


Fig. 5 Variations in ammonia nitrogen content regarding different oxygen concentrations. [Time(d), Ammonia nitrogen]

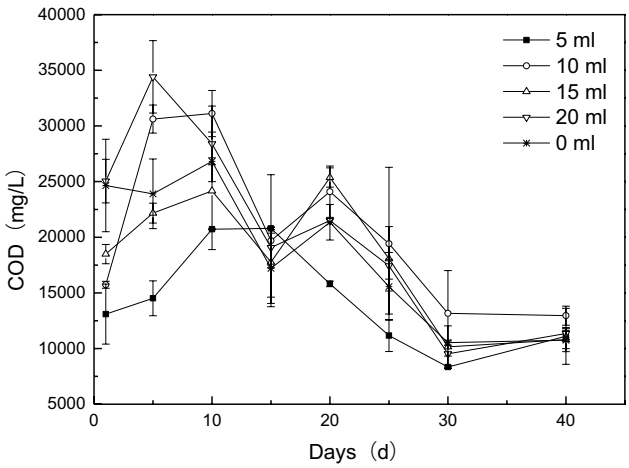


Fig. 6 Variation in COD content under different oxygen concentrations. [Time(d), COD]

molecules such as VFAs which lead to an increase in gas production, consistent with our earlier analysis. Therefore, the content of small organic molecules decreased, followed by a decrease in the COD. In terms of the overall COD, the micro-oxygen pretreatment with an appropriate oxygen concentration could promote the anaerobic fermentation progress. The inhibition of anaerobes by a small amount of oxygen was more robust than the promotion of facultative anaerobes, causing the decline in overall hydrolytic efficiency of the system. Feng et al. investigated the mechanism of micro-oxygen pretreatment on anaerobic fermentation. The results show that with the addition of O_2, the cumulative gas production, COD and other parameters are improved. It is indicated that micro-oxygen activates aerobic and facultative anaerobic microorganisms in the system and promotes its participation in the hydrolysis and acidification process of macromolecular organic matter [40]; Other studies have shown that the addition of O_2 can stimulate the chemical corrosion of sulfur-oxidizing bacteria and iron to induce microbial oxidation of sulfides to improve the methanogenic efficiency and control the H_2S concentration in biogas [41]. After comparison it was observed that R4 (with 20 mL oxygen supply) had the highest peak value, and it reached the peak the earliest, thus demonstrating that, without interfering with the anaerobic environment, oxygen supply of 20 mL could boost the activities of facultative anaerobes, enhance the decomposition of OM, improve the gas production efficiency of anaerobic fermentation, and shorten the duration of the anaerobic fermentation cycle. In the present study, optimal hydrolysis and gas production was observed in the R4 (20 mL oxygen supply) system.

Conclusions

The daily methane yield displayed different patterns at different concentrations of oxygen in the anaerobic fermentation of kitchen waste. From the gas peak value and the time required to reach the peak, it can be inferred that at oxygen supply lower than 15 mL which is 0.83 mL/(gVS•d), the inhibitory effect on anaerobes was far stronger than the promotive effect on facultative anaerobes. The system exhibited significantly reduced gas production during the entire later stage of fermentation. The treatment group R4 exhibited the highest gas production with a cumulative methane yield of 399.25 mL/gVS, and reached the peak the earliest, thereby shortening the duration of the anaerobic fermentation cycle.

Compared to the control group R5, groups with different micro-oxygen concentrations demonstrated varying COD. The treatment group R4 with 20 mL oxygen supply presented the highest ammonia nitrogen and COD during the digestion, with an ammonia nitrogen content of approximately 500 mg/L and a stable peak COD value of 34410 ± 510 mg/L. The results demonstrated that an oxygen supply of 20 mL, which is 1.11 mL/(gVS•d) could promote microbial utilization and decomposition of nitrogenous molecules, as well as other small organic molecules, improving the efficiency of anaerobic fermentation.

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